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## Sketch-based interactive visualization: a survey

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**Abstract** Interactive visualization has been an active subfield of scientific visualization for a longtime, in which user interfaces have traditionally followed the WIMP (Window, Icon, Menu, Pointer) paradigm. Though functional and powerful, they can also be cumbersome and daunting to a novice user, and exploring complex data requires considerable expertise and effort. A recent trend is toward more easy-to-use and *natural user interfaces*, which has led to interfaces like sketch-based one for interactive visualization (SIV). The goal is to allow intuitive interactive visualization by sketching in the visualization process, from data filtering to rendering. In this state-of-the-art report, we give an overview of relevant research works related to sketch-based interface in visualization. We discuss how sketch-based interaction takes effect at specific stages of the visualization pipeline. We present a categorization based on the aim of interaction, of which there are seven primary categories: selection, cutting, segmentation, matching, coloring, augmentation, and illustration. What is more, we present important items related to SIV interface design, including SIV system modes, necessary tools and some fundamental principles. This survey also provides an overview of some specific applications of SIV and a discussion of important challenges and open problems for researchers to tackle in the coming years.

**Keywords** Interactive visualization · Sketch-based interface · Natural user interfaces · Sketch-based visualization

### 1 Introduction

In the 1980s, with the advent of scientific computing and the increasing computing power, visualization, as a method of computing McCormick et al. (1987), was born to deal with the expanding scientific data. Aims to let users see the unseen by transform the symbolic into the geometric, visualization has been playing important role in medical imaging, computational fluid dynamic (CFD) simulation, meteorology, biology, geology, earth and space sciences, and many other fields.

Card et al. defined visualization as “The use of computer-generated, interactive, visual representations of data to amplify cognition” (Card et al. 1999). Visualization appears to have two main components: representation and interaction. The latter has been widely regarded as one of the top scientific visualization research problems for many years (Sutherland et al. 1966; Hibbard 1999; Tory and Moller 2004; Johnson 2004; Johnson et al. 2006). And some close related terminologies are put forward, for example, interactive visualization Müller and Schumann (2002), interactive visual analysis (Konyha et al. 2009) and related

visual analytics (Thomas and Cook 2005) etc. Laramee and Kosara (2007) presented a selected list of future challenges and top unsolved problems in visualization with an emphasis on human-centered visualization.

The well-known visual-information-seeking mantra: “Overview first, zoom and filter, then details-on-demand” by Shneiderman (1996) elucidates the role of interaction and navigation in visualization design. Since then researchers have brought forward some interaction mantras (such as Keim et al. 2006), all of which emphasizes the role of interaction in visualization while prompts general visualization pipeline for researchers and engineers.

Generally, one of human computer interaction’s (HCI) targets is to fill the gap between human and computer, which holds true for HCI in visualization. In visualization researchers’ words,

How could user handle with the data via visualization *intuitively*?

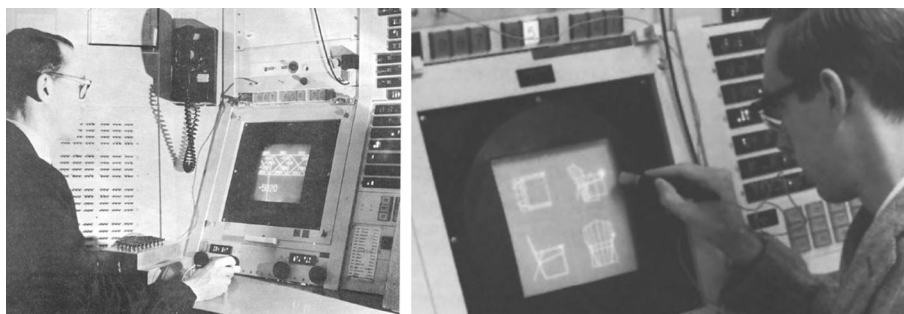
The answer of the question above is twofold at least: intuitive interaction and visual feedback on the fly. “Good interactive tools should not draw attention to the novelty of their operation. Instead, they should just seem natural and obvious, bolstering and never confounding human cognitive capacity” (Pike et al. 2009). Sketch-based interaction, as a distinct form of drawing, provides one such technique (Buxton 2010). Sketching is a powerful tool to help make abstract ideas more concrete, since a sketch can be made quickly and simply with the tools at hand. What is more, sketch-based interaction allows users to operate on the image space. Users can directly translate their intentions to strokes on image easily. For example, if one wants to cut the result arbitrarily, he can draw a free-form stroke on the exact place to cut and the results will be updated interactively. By contrast, traditional popular visualization systems such as Tecplot (2013), EnSight (2013), VisIt (2013) and ParaView (2013), typically employ the WIMP (Window, Icon, Menu, Pointer) interface paradigm, which is based on selecting operations from menus and floating palettes, entering parameters in dialog boxes. Users have to translate their intentions to the “system language”, which is always different with different visualization systems, and then they need to find where to get there functions and how to use them. At last, the results displayed may be far from the users’ original intention.

Sketch-based interaction is used in various applications, such as image editing, 3D modeling and non-photorealistic rendering (NPR) effects design, as it is intuitive for users to achieve desired results, such as Guo et al. (2011). Sutherland’s SketchPad system (Sutherland 1964) which used a light-pen input device to directly create and manipulate on-screen objects is recognized as the first sketch-based interfaces (SBI), as shown in Fig. 1 (Sutherland 1963; Andrews 2013), following with many sketch-based 3D modeling systems, for example, SKETCH (Zelevnik et al. 2007), Teddy (Igarashi et al. 1999), Chateau (Igarashi and Hughes 2001), BlobMaker (Araujo and Blobmaker 2003) and ShapeShop (Schmidt et al. 2006; Schmidt and Singh 2008). The SketchPad system anticipated many challenges that SBI would encounter in the future, including how to accept and process user input, interpret that input as an objector operation, and represent the resulting object (Olsen et al. 2008). In addition, the Teddy system (Igarashi et al. 1999) pioneered the free-form sketch-based modeling concept. All there sketch-based modeling systems provide good references for introducing SBI into visualization, namely sketch-based interactive visualization.

At first, we try to answer one fundamental question:

What is *Sketch-based interactive visualization*?

Sketch-based interactive visualization (SIV) is one type of interactive visualization, which uses direct free-form stroke(s) to manipulate data. It is more than just visualization and can rather be seen as an integrated approach combining visualization, human factors and special knowledge. One of the main challenges in SIV is sketch interpretation as that in sketch-based modeling. The interpretation could be divided into some primary methods, such as to modify color mapping, to segment different parts (define



**Fig. 1** Ivan Sutherland’s SketchPad system (Sutherland 1963; Andrews 2013). (Images generated by I. E. Sutherland)

contour in 2D or isosurface in 3D), to cut the geometry visualized, to add details to an existing model, or to deform and manipulate a model (Sect. 3).

In this paper, we survey the state of SIV. This survey is organized as follows. After briefly discussing sketch acquisition and filtering (Sect. 2), sketch interpretation is described in detail in Sect. 3. Then we study some typical interface design issues (Sect. 4) and applications in SIV (Sect. 5). The challenges and open problems are discussed in Sect. 6, before the conclusions are drawn in Sect. 7. It is necessary to note that this paper does not aim to give a complete list of all the related papers, but gives an inspiration to this non-trial interaction method in visualization, and inspire more solutions to the first question in this survey.

## 2 Pipeline

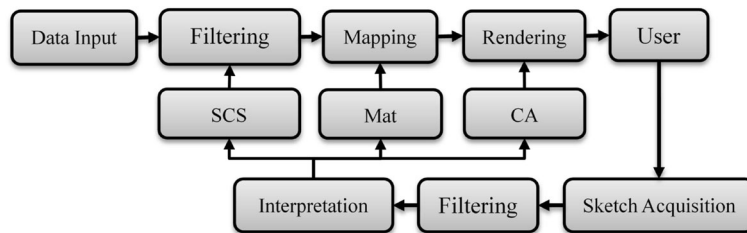
Visualization is typically partitioned into different steps, such as Hansen and Johnson (2005); Fuchs and Hauser (2009). These steps consist of the visualization pipeline which describes how visual representations of data can be obtained following a procedure of well-defined steps (Hansen and Johnson 2005). We take one admitted visualization pipeline (Hansen and Johnson 2005) for assisting to study different sketch interpretation in visualization as illustrated in Sect. 2. We found it is clear to categorize interpretation of sketching in visualization by different step in the pipeline as shown in Fig. 2. The data input step encompasses the measurement (such as sensor in meteorology) or computation (such as CFD) of scientific data. In the Filtering step, generally speaking, input data objects are used to produce new output data objects. The mapping step transforms and renders geometry in conjunction with a lookup table. In the rendering stage, the information inside the data and the corresponding representations are translated into an image and displayed for the user.

On the other hand, besides existing visualization pipeline, the sketch-based interaction has three steps as shown in Fig. 2. The first stage is to acquire a sketch from the user (Sect. 2.1), followed by a filtering stage to clean and transform the sketch (Sect. 2.2). In the final stage of the pipeline, the sketch is interpreted as the specification of or operation on a 2D image or 3D geometric (Sect. 3).

### 2.1 Sketch acquisition

Sketch acquisition is the fundamental step that obtains user input stroke(s) by variable devices. Figure 3 lists some input devices have been used in SIV. As a widely used computer input devices, 2D mouse (Fig. 3a) is the most significant medium for computer input. When designing SIV system we should take consideration on how to be useful with mouse only. The second common used HCI device is the pen and tablet (such as Cintiq Wacom 2013 in Fig. 3b). Compared with the mouse, the straightforward of pen and tablet is greatly fitful to sketch-based interaction, because sketch-based interaction essentially mimics drawing on paper with pen. These 2D input devices sometimes require mapping the user's 2D movement in user space into 3D coordinates in the virtual space, but not as frequent as in sketch-based modeling.

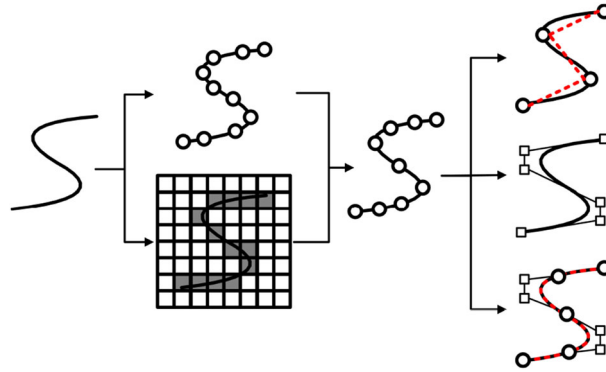
As the name hints, 3D mouse, such as SpaceMouse 3Dconnexion (2013) shown in Fig. 3c, is used to interact with computer in three dimension. It gives users more freedom but needs specific computation. Unlike traditional interfaces that provide visual and auditory information, haptic (from the Greek *haptēsthai*, meaning “to touch”) interfaces (such as Omega7 Dimension 2013 shown in Fig. 3d) generate mechanical signals that stimulate human kinesthetic and touch channels (Hayward et al. 2004), realizing tactile communication between human and computer, and requiring computation of the reaction forces at kilohertz



**Fig. 2** The SIV pipeline and SBI classification: after acquiring and filtering the input sketch, the sketch can be interpreted as an operation in 3D. *SCS* stands for Selecting, Cutting and Segmentation; *Mat* stands for Matching; *CA* stands for Coloring and Augmentation



**Fig. 3** Typical input devices: **a** Mouse (Wacom 2013) (image generated by Wacom Inc.) **b** Pen and Tablet (Wacom 2013) (image generated by Wacom Inc.) **c** 3D Mouse (3Dconnexion 2013) (image generated by 3Dconnexion Inc.) **d** Haptic device (Dimension 2013) (image generated by Force Dimension Inc.)



**Fig. 4** Sketch representation and then filtering. Circles denote ends of straight-line segments, while squares represent curve control points. Adapted from Olsen et al. (2009)

speeds. Haptic devices have been used in many areas, such as medical training and surgery, Scientific Visualization (Taylor and Russell 2005). However, 3D devices, including 3D mouse and haptic interfaces, face two critical problems: occlusion and confusion. As a result, users usually need special training.

## 2.2 Representation and filtering

Sketch representation and sketch filtering are common steps in SIV and sketch-based modeling. This section briefly introduces these steps, please reference to Olsen et al. (2009) for more detail.

As shown in Fig. 4, from left to right, once stroke is input into computer, it will be sampled or approximated with a pixel grid. The number of sample points is determined by sample speed and the length of stroke. The sample speed is different with diverse input devices. It is necessary to aware that sample points are usually not uniform distributed. Meanwhile, strokes can be approximated with a pixel grid. This method is inspired by image processing. It is made by recording the pixel grids that the input strokes pass through.

Then some necessary resampling and smoothing steps will be used to obtain uniformly sampled points without noise at times. In order to reduce the noise, resampling is executed on the fly or offline. For example, if the distance between two points is less than a threshold, the latest point is discarded; if the distance between two points is too large, we interpolate to get new point between these two points.

At last, in this phrase, approximation or fitting the sketch to other representations is sometimes required for two reasons: one is simply the input, the other is making it easier to compare against each other (Olsen et al. 2009). At the most right in the Fig. 4, from top to down, there are three typical approximation and fitting methods. The first one, called polyline approximation, uses few points connected with straight-line to represent the stroke(s). The second one, called fit-to-curve, uses Bezier or B-spline curve to fit the stroke(s). The last one integrates foregoing methods. That is using polyline to approximate part of stroke with small curvature and curve to fit other part of stroke with high curvature.

### 3 Interpretation

After sketch has been sufficiently filtered, the next key stage is interpreting the sketch. Different from sketch in modeling which normally maps strokes to various models or details of models, sketch in visualization vastly interprets as respective operations on visualized data.

In practical applications, to improve the analysis process, it is often desired to not only render the data but also to interactively edit the data. Examples thereof include the manual classification and segmentation of structures, the removal of structures to uncover regions of interest and thus to isolate important parts of the data, or the coloring of parts to emphasize relevant structures and to give extra information about them. Such mechanisms can help to effectively reveal and communicate the relevant information in 3D scalar and vector fields and to create images that are easy to understand even by novices.

As in sketch-based modeling, one key issue in designing SIV systems is how to infer missing information (e.g. depth for 2D input devices). The visualization systems need to understand and interpret sketches in three dimension in lots of cases, such as sketching on 3D visualization results. And it is helpful to learn from sketch-based modeling (Olsen et al. 2008, 2009; Cook and Agah 2009). In this section, we discuss interpretation of sketch in detail based on classification illustrated in Fig. 2. It is necessary to notice that some of the interpretations are related with general “gesture recognition” (Wigdor and Wixon 2011) problem. However, we discuss these interpretations in specific visualization context.

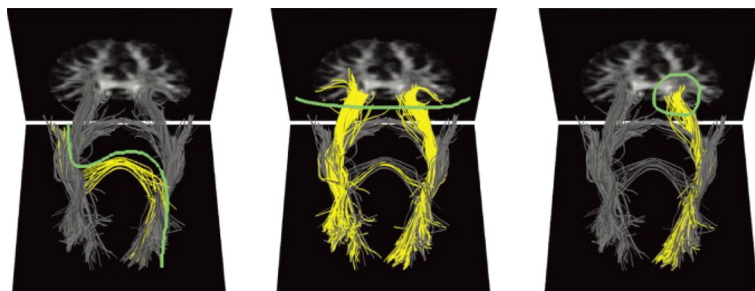
#### 3.1 Selecting

We can take advantage of the properties of sketching to select data. In particular, we could use the region separation property to select data. Selecting or extracting data with a stroke means choosing cells and points (and associated attribute data) that lie within in the region defined by the stroke or intersect with stroke(s). To determine whether a point  $x$ - $y$ - $z$  lies within a region or intersect with stroke(s), one could simply evaluate the point and examine the sign of the result. A cell lies in a region if all its points lie in the region, and a line intersects with stroke(s) if one stroke pass through the line in image space or 3D geometry space.

Based on VolumeShop Bruckner and Groller (2005), Pühringer (2009) realized a method for selecting specific structures of interest in a volume dataset, which cannot be separated by a transfer function setup. They generate 3D models of 2D input strokes, and then fit the generated model on volume structures. Users can pick a structure of interest within a volume dataset with 2D sketches smoothly without disruption.

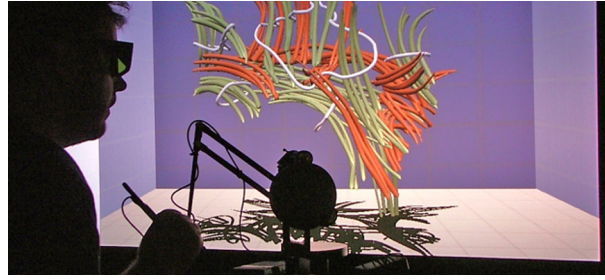
In CINCH Akers (2006a, b) described a 3D pathway selection interface. By a cooperating prototyping approach with special scientists, three significant marking operations were invented: shape match (selects similar pathways) using mean closet points metric in image space; touch mark (select touched pathways) based on the sweep surface Owada et al. (2005); surface intersection (select pathways that intersect a specified region on a cutting plane) inspired by Olwal et al. (2003) and growing and shrinking selections (Fig. 5). Another important contribution of this work is the interface design principles resulted from the participatory design process, namely interactivity, minimality and visibility. It is worth to note that CINCH was developed from their previous works Akers et al. (2004); Sherbondy et al. (2005) which used ROI-based selection techniques. In a word, this work provides a good paradigm for developing practical interactive visualization interfaces.

In Keefe et al. (2008), proposed a virtual reality (VR) technique for input of smooth 3D trajectories with varying curvature, called dynamic dragging. The key in this technique is determining the best mapping from

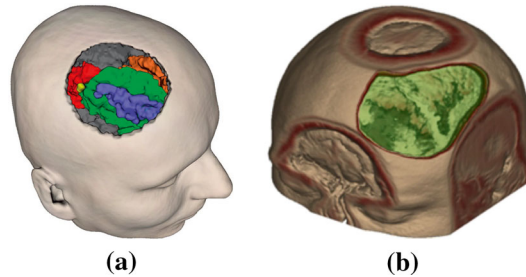


**Fig. 5** Users of CINCH invented three marking operations during participatory design. *Left* shape math mark, *Middle* touch mark, *Right* surface intersection mark Olwal et al. (2003). (Images generated by A. Olwal)





**Fig. 6** Users can select objects with variable curvature strokes (Keefe 2010). (Image generated by D. F. Keefe)



**Fig. 7** Illustrations of typical cutting operation: **a** Parts of bone isosurface in an MRI data set were removed with free-form strokes to reveal interior brain structures (Bruckner and Groller 2005). **b** Interactive cut and sketch (Chen et al. 2008). (Images generated by H.-L. Chen)

the input encountered thus far to the dragline length. The authors took three strategies, namely local curvature, drawing speed and integration of the first two. With 3D input device, users can select objects with variable curvature strokes, see Fig. 6.

Yu et al. (2012) presented free form point cloud selection methods, including TeddySelection and CloudLasso. For both techniques, the interaction is based on sketching on the 2D projection of the 3D space.

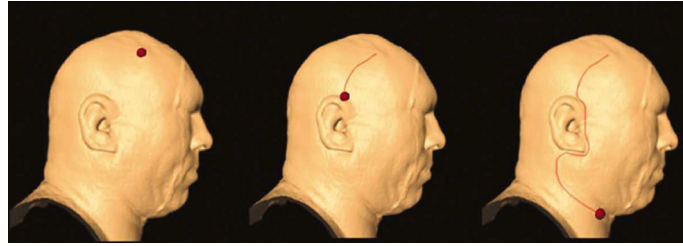
### 3.2 Cutting

We often want to cut through a dataset with a surface and then display the interpolated data values on the surface. We refer to this technique as data cutting or simply cutting. The data cutting operation requires two pieces of information: a definition for the surface and a dataset to cut. The surface can be defined with a stroke. A typical application of cutting is to slice through a dataset with a plane, and color map the scalar data and/or warp the plane according to vector values. We can view data on arbitrary cutting surfaces with strokes.

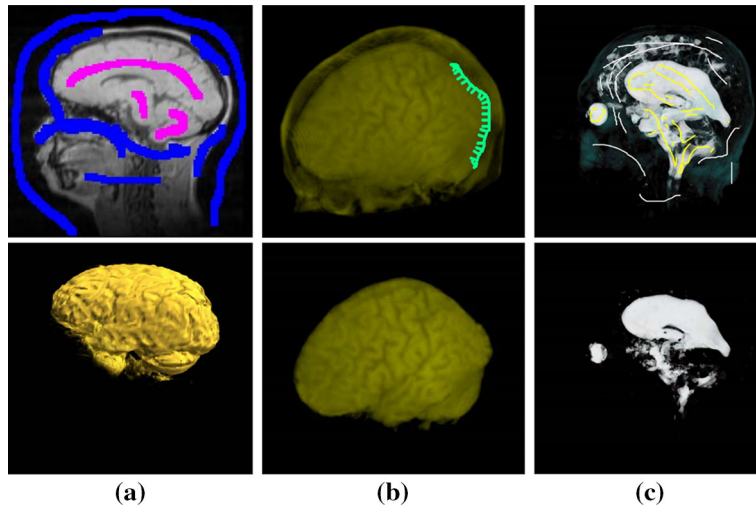
Based on VolumeShop Bruckner and Groller (2005), Burger et al. (2008) present an efficient graphical processing unit (GPU) realization of the volume painting method. They also extended on the idea of selection volumes and present a volume editing technique to explore internal structures. As shown in Fig. 7a, users sketch on the bone isosurface to remove any parts to reveal interior brain structures. It is noticeable that this editing technique is not limited to volume resolution.

Chen et al. (2008) proposed a GPU-based point radiation technique to support real time direct drilling, lasering, peeling, and cutting/pasting. The key idea is to create a set of 3D points associated with the existing voxels assuming that each point has a continuous field of energy (radiation) that drops smoothly off to zero. More importantly, they enable parallel region growing segmentation that allows multiple seed planting by direct sketching on different volumetric regions (Fig. 7b).

Bruyns and Senger (2001) developed a method simulate cutting, removal and realignment of tissue, allows the user to directly sketch the desired cut contour on a three-dimensional surface like dragging scissors through fabric (Fig. 8). The entire cut procedure contains four steps: placement (position where cutting begins), activation (start cutting), cutting (advance the cutter's position), and deactivation (finish cutting operation).



**Fig. 8** Tissue cutting (Bruyns and Senger 2001). (Images generated by C. D. Bruyns)



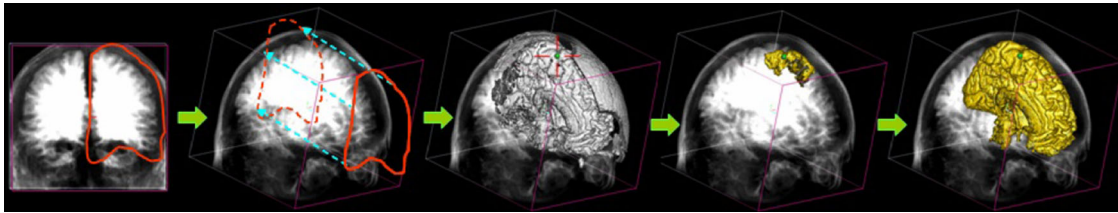
**Fig. 9** Sketch-based volume segmentation: **a** *Top* A painted slice, *Down* the segmented volume (Tzeng et al. 2003). **b** *Top* Drawing a 2D stroke along the contour of the brain, *Down* Result 3D region (Owada et al. 2005). **c** *Top* Define foreground and background with freehand strokes, *Down* the result volume (Yuan et al. 2005). (Images generated by X.-R. Yuan)

### 3.3 Segmentation

Segmentation is the process of splitting data into different perceptual or semantic parts. As an extensive research area, typically in medical, segmentation algorithms range from very general methods to specifically designed to identify certain structures. The ratio between correctly and incorrectly classified voxels is usually used to evaluate different segmentation algorithms. In practice, this criterion is difficult to measure because of limited information while there are many sophisticated segmentation algorithms proposed. However, automatic segmentation algorithms could not divide distinct unions successfully due to high-level recognition related to the semantics of data. To overcome the difficulty of automatic segmentation, human supervision or intervention becomes essential.

Tzeng et al. (2003) developed an intuitive user interface for specifying the classification functions that consists of users painting directly on sample slices of the volume. The system allows user to sketch on the cross-section of volume data that roughly indicate foreground and background regions (Fig. 9a). Then the points in painted regions are sampled as the input of the neural network, which calculates a set of weights for volume classification. In Tzeng et al. (2005), they extend their method in ways, such as instead neural networks-based classifiers with machine learning classifiers.

In Volume Catcher Owada et al. (2005), to release users from drawing too much strokes, Owada et al. described a method to segment volume directly by 2D free-form sketching on the rendered image. Assuming the user always traced the silhouette of the object of interest, the system computes the 3D locations of the user-specified 2D strokes by projecting a user's mark into a sweep surface, and returns a volumetric region inside the stroke by performing segmentation inside the stroke (Fig. 9b). Later, the authors optimized the system by pre-process user's input in 2D domain before applying the traditional 2D–3D stroke elevation



**Fig. 10** A seeded region growing volume segmentation system (Chen et al. 2006). (Images generated by H.-L. Chen)

algorithm (Owada et al. 2008). What is more, the optimized system combined with many more user interface tools: Contour, Click, Axis, Grabcut, and Foreground/Background.

Inspired by the Volume Catcher method (Owada et al. 2005), Yuan et al. (2005) proposed a method for cutting out 3D volumetric structures based on simple strokes drawn directly on rendered images without limitation of specific location (Fig. 9c). Their system automatically segments the regions of interest according to the freehand strokes, and then boundaries of the regions were refined with 2D graph cuts algorithm. They accelerate the whole process by pre-computing over-segmentation.

Similar to the Volume Catcher method (Owada et al. 2005), Chen et al. (2006) introduced seeded region growing volume segmentation system (Fig. 10). The system extends the stroke and sweeps through the volume, so users do not need to browse through hundreds of cross-sectional slices. The process can be outlined as follows: Initially, the user specifies a region of interest by placing a closed free-form sketch on the volume rendering. Then a region of interest is extruded to facilitate the definition of seed points for the region growing. However, selecting from hundreds of slices and pinpointing the correct location on the 2D image require both highly trained personnel and time commitment.

As a natural extension to color mapping, contouring (isosurface in 3D) construct the boundary between different regions, such as 2D isotherms and constant elevation in meteorology, 3D isosurfaces with constant intensity corresponding to different body tissues in medical. With VolumeViewer (Sowell et al. 2009), developed an interface, for modeling surfaces from volume data based on sketched contours on arbitrarily oriented cross-sections of the media volume data (Fig. 11). They made use of the surface reconstruction algorithms in Liu et al. (2008), which introduced a new surface reconstruction from non-parallel curve network. The interface provided three sketching tools: paint, dots, and LiveWire.

### 3.4 Matching

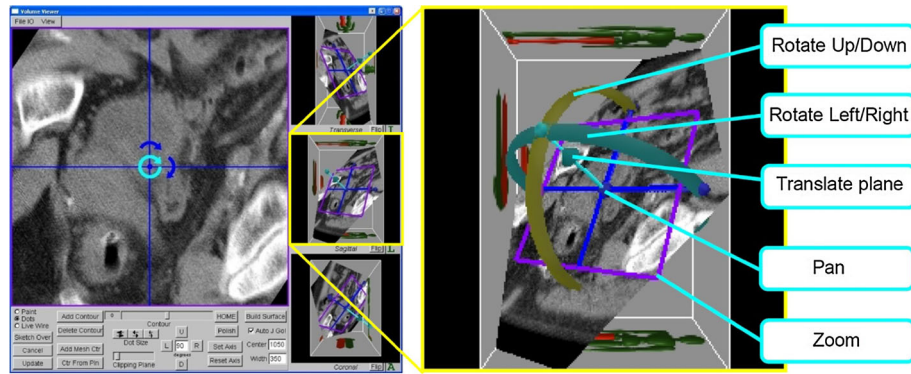
Matching selects geometries like the one illustrated by stoke. Field line (such as streamline, pathline, streakline and timeline) matching is to find specific shape of field line similar to sketched strokes. This type of interaction with visualized data is hard achieved by traditional methods. Matching can not only find shapes of interest, but also alleviate some intractable problems. For example, streamline seed point problem, which is very important in flow visualization (Post et al. 2003; Laramée et al. 2004). Several algorithms for automatic seed point placement (Rosanwo et al. 2009; Li et al. 2008; Marchesin et al. 2010) and interactive seed point placement (Bryson and Levit 1992; Schirski et al. 2003; Bürger et al. 2007) have been presented.

Wei et al. (2010) proposed a user-centric approach to exploring 3D vector fields. Their method allows the user to sketch 2D curves for pattern matching in 2D and field lines clustering in 3D (Fig. 12a). More importantly, they employed an automatic clustering method to generate field-line templates for the user to locate sub-fields of interest (Fig. 12b). The system utilizes string matching approach to find the field line (which has being mapped to 2D and approximated by an arc length parameterized cubic B-spline) that is most similar to the input. However, this system assumed that enough field lines that are sufficient to capture as many features as possible in the vector field have been computed. This is hardly time-consuming, especially for large scientific data.

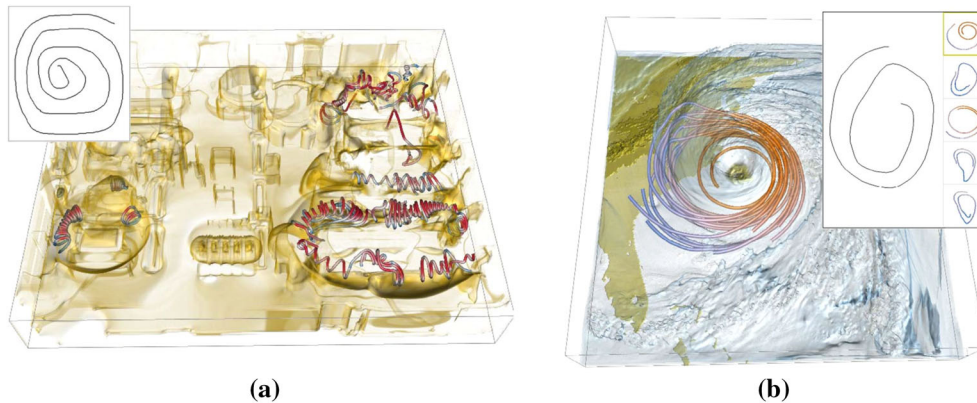
### 3.5 Coloring

Coloring (Color mapping) as a common scalar visualization technique maps scalar data to colors and displays the colors using the standard coloring and shading facilities of the graphics library. The scalar mapping is implanted by indexing into a color lookup table. Scalar values serve as indices into the lookup table. The key to color mapping for scalar visualization is to choose the lookup table entries carefully.

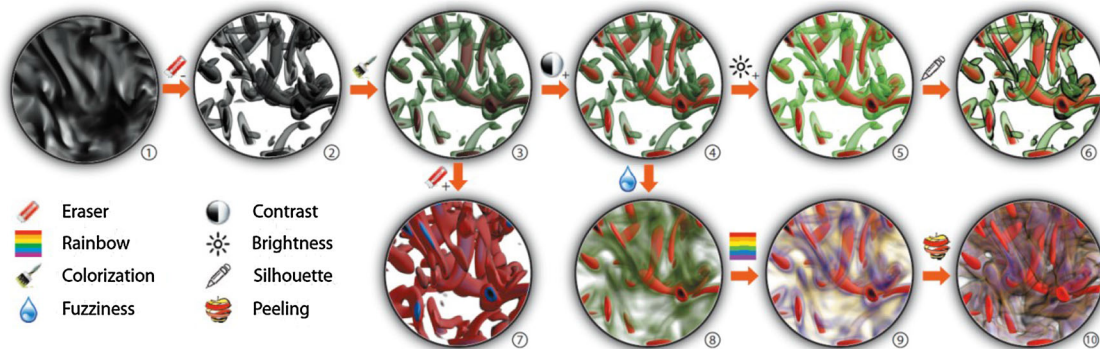




**Fig. 11** The VolumeViewer Interface (Sowell et al. 2009). (Image generated by R. Sowell)



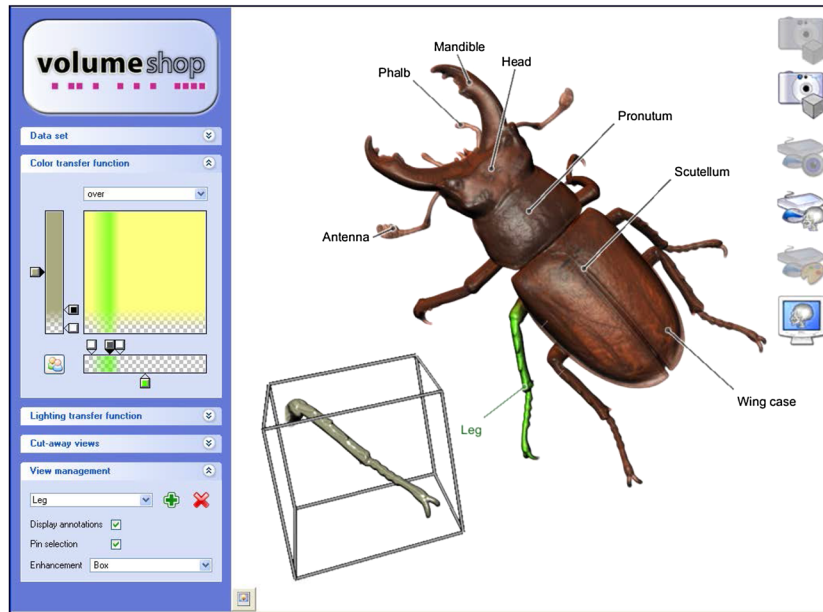
**Fig. 12** Left sketch-based clustering, Right sketch-based template query (Wei et al. 2010). (Images generated by J.-S. Wei)



**Fig. 13** The WYSIWYG volume visualization system comprises several useful tools for sketch-based visualization (Guo et al. 2011). (Images generated by H.-Q. Guo)

Traditionally, users generally need to specify colors of different data values by widgets. This process is nonintuitive and time-consuming. What is more, any type of lookup table can exaggerate unimportant details or create visual artifacts because of unforeseen interactions among data, color choice, and human physiology (Hansen and Johnson 2005). By sketching on the displayed image(s), users can directly and clearly modify colors of certain parts, whose process is comparably intuitive and alleviates even settle the unexpected visual artifacts, therefore sketching is extremely helpful for uses, especially novice.

In practice, lookup tables, generally called transfer function (Kindlmann 2002), are designed by the users to accentuate important features while minimizing less important or extraneous details. However, what are



**Fig. 14** VolumeShop (Bruckner and Groller 2005). (Image generated by S. Bruckner)

important features? The answers are different in certain degree even though there are some common characteristics for general features. By sketching, user can also clearly sketch features that important for himself.

With VolumeShop (Bruckner and Groller 2005), proposed an interactive system for direct volume illustration (Fig. 14b). The system creates a fully dynamic three-dimensional illustration environment, which directly operates on volume data. It discriminates between two basic types of volumes: data volumes and selection volumes, the latter specifies a particular structure of interest in a corresponding data volume. With simple sketching, specific features will be displayed without occlusion.

Wu and Qu (2007) presented an image-centric (Pfister et al. 2001) method for editing direct volume rendered images (DVRI). Their approach allows users to fuse multiple features in different DVRI into a comprehensive one, to blend two DVRI, and/or to delete features in a DVRI with sketching. The key idea is to transform the DVRI editing problem to an optimization problem with an energy function (based on the image similarity values between the target DVRI and the source DVRI) created for each editing operation, and then Genetic Algorithms are used to search out good transfer function for fusing multiple features.

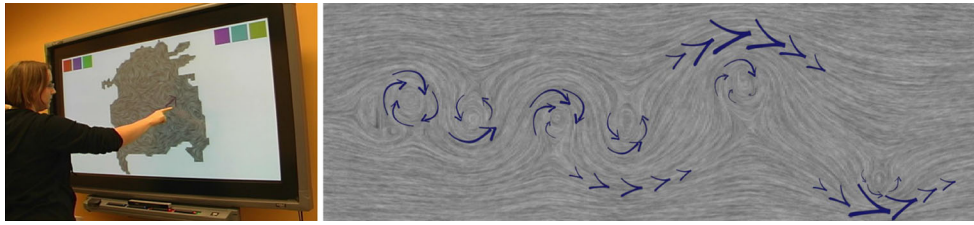
Guo et al. (2011) represented a volume visualization system that accepts direct manipulation through a sketch-based What You See Is What You Get (WYSIWYG) approach. Their system developed a set of operations, such as painting different colors, changing opacities, erasing volume structures, modifying contrast and brightness, adding silhouettes, peeling volume feature layers and so on (Fig. 13). Their tools can be easily reached by brushing a few strokes on top of the rendered volume image.

Ropinski et al. (2008) present a user interface for the design of 1D transfer functions. Based on the stroke(s) drew directly onto the volume rendering near the silhouette, the system performs a histogram analysis to identify the desired feature in histogram space, and then a component transfer function is automatically generated. As a result, different regions in the data are identified and rendered.

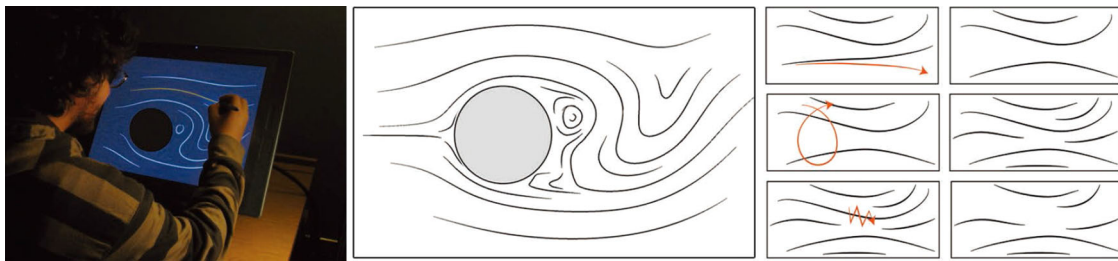
### 3.6 Augmentation

Augmentation is to append symbols to annotate or augment visualization result. This is useful to emphasize important features in the data, enhance the perception of structures in the data and the relationships between them. Such augmentation is very useful in collaborative visualization, interactive exploration and education.

Isenberg et al. (2008) present techniques to interactively sketch custom-design glyphs that reveal patterns of the underlying dataset with hand postures in 2D (Fig. 15). Users could easily modify glyph



**Fig. 15** Hand-sketched glyphs are used to point out aspects of the flow (Isenberg et al. 2008). (Images generated by T. Isenberg)



**Fig. 16** Drawing with the flow to illustrate simulated 2D fluid flow past a cylinder (Keefe and Isenberg 2013). (Images generated by D. F. Keefe)

placement, glyph density and shape. What is more, the interface of this system is controlled through four hand postures, which represent different operations on the flow field visualized with LIC algorithm.

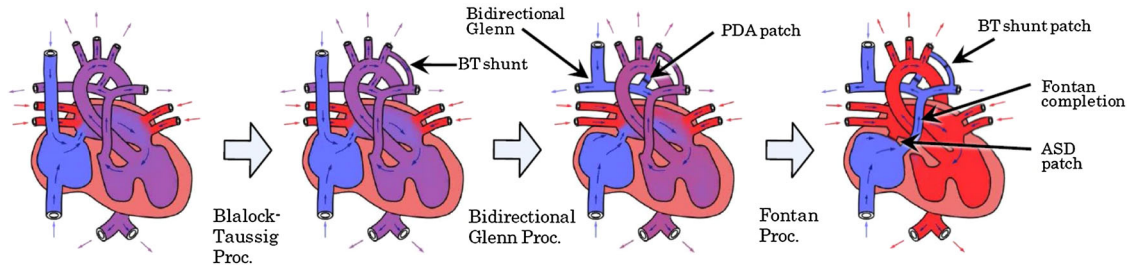
Besides adding glyph into visualization, we can also append appropriate text illustrator. For instance, in VolumeShop (Bruckner and Groller 2005), the authors added annotated illustration to a human foot. While in Burger et al. (2008), added special glossary to a human skull with different color, inspired by textbook. And they extend their idea to add pertaining and surface-aligned annotations to diverse objects.

### 3.7 Illustration

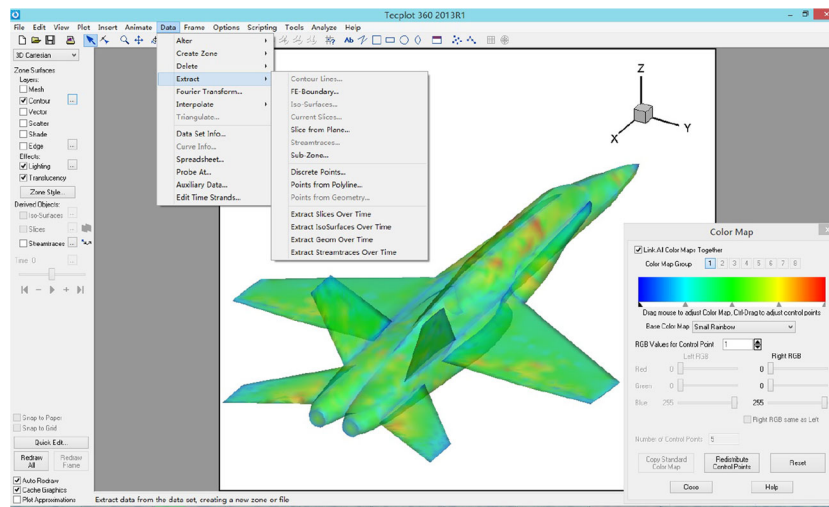
Illustration visualization refers to a computer supported interactive and expressive visual abstraction motivated by traditional illustrations (Rautek et al. 2008). It plays an important role in areas like education process. It is an essential tool in communicating complex relationships and procedures in science and technology. In medical, such as surgical procedure teaching, normal or aberrant anatomy or heart can be illustrated, and some medical device can be explained how to use in practical. In CFD, professionals could describe physical phenomena. The basic goal of illustrative visualization is to enhance the perception of structures in the data and the relationships between them by emphasizing important features.

Schroeder et al. (2010) presented a sketch-based system for making illustrative visualization of 2D vector fields accessible to illustrators (Fig. 17). Mostly the strokes drawn by the illustrator may not exactly match the underlying vector field data. The system reconciles these differences through a procedure called ink-data settling. That is done by resampling a set of even distributed points on the stroke, then streamline which passes through one of these points and has the same length as the original stroke is calculated. At last, the best-matched streamline is filtered by computing a similarity metric. More importantly, the system supports three gestures, such as crop, more and delete, to optimize resulting streamline distribution.

Another interesting use of sketch-based interaction in visualization is Virtual Reality (VR) visualization design. Keefe et al. (2008) proposed a methodology called Scientific Sketching. The method investigates tools methodologies for artist-scientist-technologist collaboration in designing multivariate VR visualizations via four studies, which comprise CavePainting visualizations, data-driven glyph sketching, appropriate artistic control for science, and design via scientific sketching. They got that “effective artist-accessible design tools must support quick sketch-like visual exploration, sufficient control over form to represent complex subjects in science, and an ability to capture the animated views and interactive scenarios that are critical to VR experiences”.



**Fig. 17** A sketching system that incorporates a back-ground fluid simulation for illustration of complex fluid system (Zhu et al. 2011). (Images generated by B. Zhu)



**Fig. 18** Tecplot Popular interactive visualization system with typical WIMP paradigm Tecplot (2013). (Image generated by Tecplot Inc.)

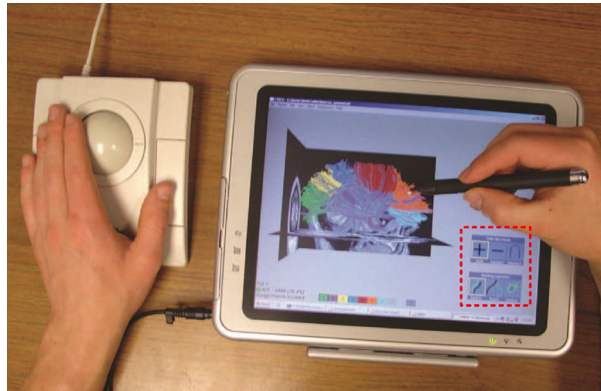
Zhu et al. (2011) presented a sketching system that incorporates a back-ground fluid simulation for illustration of complex fluid system. Users can sketch on a multilayered two-dimensional canvas to design the shapes and connections of a fluid circuit. Then the system automatically analyzes these input sketches and abstract into a hydraulic graph. This system has two notable features: one is the intuitive sketching interface, which is simple and easy to use; the other is real-time flow simulation combining a hydraulic network model and a multilayered hydrodynamics model.

#### 4 Interface design

A complete visualization system must simultaneously support many operations such as data read in, view and rendering controls, color and opacity control and different representation type. A traditional visualization system makes use of WIMP interface paradigm. For instance, Tecplot (2013) is one of the most widely used CFD post-processing software. As shown in Fig. 18, it provides abundant functions by menus and wizards etc. However, this type of interface paradigm is not direct for data manipulation and hard for novice.

Most SIV systems provide a prototype application with useful interfaces for visualization and interaction (Bruckner and Groller 2005; Ropinski et al. 2008; Zhu et al. 2011; Keefe et al. 2008; Keefe and Adviser-Laidlaw 2007; Sowell et al. 2009), lots of which are inspired by image processing software like Adobe Photoshop (Photoshop 2013) or geographical software like Google Maps (Map 2013). At most time, SIV interfaces contain some kinds of mode. Different kinds of mode means different interpretation operations, such as selecting, cutting, segmentation, coloring and augmentation as discussed in Sect. 3.





**Fig. 19** Explicit mode switching in SIV system, highlighted with *red dashed rectangle*. Reproduced from Akers 2006a, b. (Image generated by D. Akers)



**Fig. 20** Implicit mode switching in SIV system. Reproduced from Isenberg et al. (2008). (Images generated by T. Isenberg)

#### 4.1 SIV system modes

There are two manners to switch between different modes: explicit and implicit. The former, namely explicit switching, provides visible widget to allow user choose one desired mode, such as Guo et al. (2011); Zhu et al. (2011); Olwal et al. (2003). For instance, Fig. 19 illustrates explicit mode switching in SIV system, highlighted with red dashed rectangle. This manner's strongpoint is clearly revealing the current interaction mode. Meanwhile, the weakness is interrupting users' analyzing process frequently.

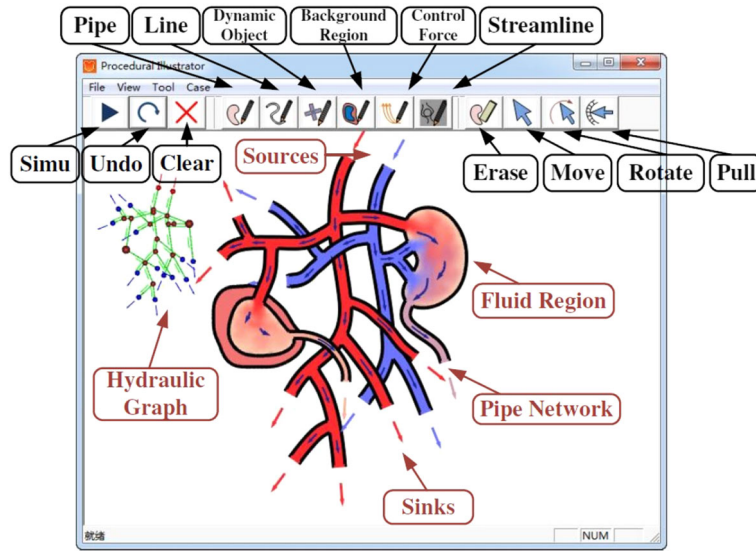
The latter, namely implicit switching, conjectures users intention by kinds of information got from the procedure of sketching (Guimbreti re and Winograd 2000; Hinckley et al. 2005). Some systems use different shapes of strokes to indicate diverse modes, like Owada et al. (2008). Some systems use different gestures to represent several modes, for example, Isenberg et al. (2008) introduced a system supports different gestures, such as one figure, two figures, fist and flat hand, shown in Fig. 20. This manner supplies fluid interaction, while needs special training and inclines to mistake.

A good SIV system should integrates there two manners to provide fluid interaction, to release users from frequent switching between different modes, and to reduce learn and memory burden. What is more, if there are more than one induced interpretation according to the stroke drawn by the user, it is necessary to prompt up there candidates. As such, the user can further confirm one desired operation. At last, mode clue is necessary to tell the user in which mode they are.

#### 4.2 Necessary tools

SIV systems should provide users with various necessary tools to help them handle and edit these visualization results. Without rotation, translation and zooming (Cockburn et al. 2008), the system could support interactive change color and opacity, for example, the system in (Guo et al. 2011) allows users sketch on the visualization result to change colors of selected regions interactively. Undo and Redo are important





**Fig. 21** Illustration of necessary tools in SIV system (Zhu et al. 2011). (Image generated by B. Zhu)

functions for system, especially SIV system, since the drawn stroke is not always consistent with the user's intent due to subjective and objective factors, such as device error, as shown in Fig. 21.

Another necessary tool is drag-and-drop, which allows user easily drag objects selected with strokes to distinct position and so on. Additionally, it is very helpful if the system provides the visual histories of users' operations, so that users could simply know the phases they have. In addition, linked with undo operation, users can go back any passed phases.

It is invaluable to integrate with other important interaction technologies rationally, such as overview and context (Cockburn et al. 2008), focus and context (Furnas 1986; Muigg et al. 2008), multi-linked-view (Buja et al. 1991; Blaas et al. 2007), brushing (Becker and Cleveland 1987; Doleisch and Hauser 2002). For instance, Tzeng et al. (2005) made use of multi-linked-view to give clues in different views for user drawing strokes, as shown in Fig. 15. However, there are few commercial or open source visualization systems support aforementioned interaction technologies simultaneously, including sketch-based interaction.

Some clues are important for practical SIV system, like mode clue mentioned before. Here we emphasizes other two clues: depth clue and visual clue. Depth clue avoids the user from depth confusion in 3D. It is more important in 3D interaction. Visual clue tells the user where they are, which tool they are using and some other necessary information like that.

### 4.3 Principles

There are some necessary principles that we should follow in design SIV interfaces. The first and most important one is real-time visual feedback. That is to say, a system should give users visible hint once interaction happened. This principle is applicable for interactive system, like sketch based modeling system. However, it is more critical and difficult for SIV system since visualization generally deals with larger data than other sketch-based applications. The scale even reaches PB and more. Researchers confront challenge when interacting with these variable scales data.

How SIV system could realizes real-time visual feedback? Without several usual technologies, like data compression and feature extraction (Post et al. 2003), it is helpful to arrange interactive visualization pipeline to leverage different steps. And to use faster algorithms with the bearable loss of precision is another choice to insure interaction. For example, in Zhu et al. (2011) authors took one-way model because it is much easier to compute and control than a two-way model.

The second principle in design of SIV system interface is simplicity. That is to say, the interface of SIV system should as simple as possible with the premise that it fulfills user's request of interaction. "Good interface is no interface" (means that a good interface should show not any or less WIMP to the user) may not be a strictly accurate statement, but it emphasizes that an interactive system should has intuitive interface from the view of interaction. We believe the intuitive interface contains implicit presumes that are



**Fig. 22** Some novel paradigms of interaction with VR environments (Keefe and Adviser-Laidlaw 2007). (Images generated by D. F. Keefe)

consonant with human's mental models. It could also include some novel paradigms of interaction that are necessary and easy to learn.

Taken Keefe and Adviser-Laidlaw (2007) for example, the authors developed a VR visualization system, in which there is no traditional WIMP for users. However, as shown in Fig. 22, the left is a man uses a real bucket that has tracker attached to it to drip out virtual paint. The related parameters are calculated with information that gets from the tracker, such as speed and orientation. The right shows the similar interaction paradigms that virtual strokes are drawn with a real brush.

The third principle in design SIV system interface is cooperation. That is to say, visualization researchers should cooperate with experts from areas like CFD to design special SIV system interface. In fact, the cooperation should be more than just interface design, like algorithm design, which has been out of the range of this paper. Cooperation cannot only help optimize designed interpretation of interaction, but also enlighten new paradigms of interaction. What is more, the interface design determines proposed pipeline of interaction. It greatly affects feasibility and friendliness of the system. In order to design a practical interface, we also need to collaborate with special researchers to take them into the design process.

To avoid relying solely on the intuitions of a designer, Akers (2006a, b) applied a cooperative prototyping approach Bødker and Grønbæk (1991). This allows the authors "to cooperatively explore the consequences of design choices, better understanding possible interface ambiguities and resolution" with the help of a Wizard of Oz prototype (Akers 2006a; Dahlbäck et al. 1993). In Keefe et al. (2008) authors built on successful artist-scientist collaborations, like "Renaissance teams" developed by Cox (2008), to address sketching problem in scientific visualization.

## 5 Applications

Begun with SketchPad (Schmidt and Singh 2008), SBI has been used in several areas, such as CAD, digital cartoon, multimedia. After years' development, SBI has been playing a potential role in visualization applications. This section we discuss this use in two typical fields. The first one is medicine, in which visualization is used. The other is CFD, which plays necessary role in automotive and aviation industry and so on.

### 5.1 Medicine

SBI has been used in medical visualization and analysis. By far, SBI is mainly used to segment different tissue. That is to divide three-dimensional samplings of anatomy elements from Magnetic resonance imaging (MRI) and computed tomography (CT) scanners to different organ structures. Segmentation is very important in medical, diagnostic, and treatment planning applications, like conducting treatment, such as radiating a tumor.

However, as mentioned before, because similar densities of variable body tissues induce the ambiguity between tissue boundaries, it is fundamentally difficult to automatic segment different body parts. More

importantly, the semantic gap between the high-level human perception and the low-level features that the segmentation is based on push segmentation into more sophisticated condition. To overcome these obstacles of automatic medical segmentation, it is essential to integrate human intervention (Yuan et al. 2005).

SBI provides such a good way to provide interactive segmentation. By sketching on the different tissues (Tzeng et al. 2003; Yuan et al. 2005) or silhouette between different parts (Owada et al. 2005; Chen et al. 2006), users could get better segmented data and rendered images. What is more, an effective segmentation method can also facilitate interactive data exploration with the ability to identify and segment interesting structures or objects interactively.

Many researchers have used SBI to select parts of medicine data, to cut visualization result, and to modify color-mapping function intuitively. All these uses, including segmentation, can be extended to deal with scalar data. So their applications are not limited by medicine.

## 5.2 CFD

As a concrete application area of flow visualization, visualization in CFD is very important in simulation post-processing. To date, the simulation data from CFD has reached scale of PB. The huge data visualization is considered as one critical problem in visualization. On the other hand, facing the same data, different researchers possibly have different needs. More importantly, some typical physical features have no unified accepted definition (Post et al. 2003), like vortex. These three vital problems cause error or failure of related automatic algorithms.

SBI provides a valuable approach to handle with these problems. By sketching, users can select any interested region with any shape, which not only concentrates users' attention, but also reduce the size of the data. We can also use SBI to develop a new interactive feature extraction method to get or refine features. Take Guo et al. (2011) for example, the authors use stoke-like eraser to extrude feature, like turbulent, the eye of typhoon. What is more, SBI in flow visualization has more functions that are almost impossible to get with traditional visualization system, like matching, see Sect. 3.4.

These SBI technologies could apply to general vector field. Compared with SBI application in scalar field, there are many useful works could be done to exploit SBI in vector field, like real time querying.

## 6 Challenges and open problems

After years' development, we have made great progress in SIV. However, there are many important challenges and open problems need to be solved. That is why we have not seen many SIV systems in practice. On the way to explore a practical SIV system, or even some related technologies, we should overcome several vital obstacles as follows.

### 6.1 Efficiency

Efficiency is one of the most critical challenges SIV researches confront. To achieve useful SIV, researchers need to deal with efficiency of related algorithms. Unlike some other applications of SBI, like sketch-based modeling, it is more difficult for SIV to design and carry out efficiently. As mentioned before, the main complication comes from the growing huge dataset and insufficient computing power. To satisfy the basic interface design principle, namely real-time visual feedback (Sect. 4.3), and ensure fluid interaction, SIV related researchers should think out reasonable measures.

To date, most SIV demos provide interaction technologies to visualize and explore small-scale data. While just a short time ago data size of gigabyte was considered large, terabyte and even petabyte data sizes are now available. Although their demos is far from practical SIV system, they contain lots of useful technologies and algorithms which could assist us to reach out goal. More important, there are necessary hints for developing new SIV methods.

Besides efficient SIV interaction methods, the other fundamental critical challenge is efficient algorithms, such as volume rendering, field line computation and real-time illustration with higher resolution. However, real-time volume rendering is still a sophisticated problem for larger data, even with the help of GPU. Comparably, field line computation is more difficult to reach real-time. One key reason is that the computation of field line is harder to parallelization. The real-time illustration with higher resolution becomes more critical when displaying visualization result with display wall system, like CAVE (2013).

## 6.2 Accuracy

Unlike many sketch-based modeling systems (Olsen et al. 2009), SIV system is mainly aimed at precise results, resolving ambiguity instead of embracing it. There are at least two related challenges: precision and ambiguity. Precision is very important for visualization, since the fundamental mission of visualization is visualizing the source data and revealing the essence. That is to say, the visualization result should be consistent with the input. “This problem of precise, controlled, constrained interaction remains largely unsolved for scientific visualization applications and is a critical area of focus within this research challenge” (Keefe and Isenberg 2013).

Precision is mainly influenced by two factors: accuracy of algorithm and noise. We do not discuss the accuracy of concrete algorithm here, since it is out of the scope of this paper. The source of noise here could be further divided into data and interaction. Because device error or approximation error of numerical computation (like differential Navier–Stokes equations Acheson 1990) exists widely, how to eliminate these different errors is another challenge and deserves further research.

Ambiguous is a general problem for sketch-based application, because users mostly sketch in two dimension, which is needed to map into three dimension. However, at present, many SIV avoid this problem by sketching on image space. This is efficient for elemental research, nevertheless, because most scientific data has space information in nature, it is inevitable to confront direct sketching in three dimension when we come into practical SIV system.

## 6.3 Interface

The SIV system interface is a communication channel between users and data. The interface determines the fluid of interaction. Although interface design has been studied for many years in HCI, how to provide a more natural interface in SIV is still a problem that has attracted many researchers. We are moving into a post-WIMP world and new designs are needed. Moreover, there are still many problems with SIV system, like guessing meaning of different shape of stroke (Guimbreti re and Winograd 2000).

Various devices exist in SIV system, as illustrated in Fig. 3. However, researchers confront a challenge, namely how to make pretty use of these devices to enhance interaction of the SIV system. There are at least two problems needed to settle. First, one is how to utilize one single input device sufficiently, like 2D mouse. The other is how to integrate some devices together to realize multi-channel interaction, such as bimanual interface in CINCH (Akers 2006a, b), as shown in Fig. 19. However, it is important to note that constant context switching between different input devices would be frustrating.

Many SIV interfaces try to mimic and substitute some traditional WIMP paradigm. For instance, color mapping (Sect. 3.5) uses a transfer function dialog to modify color or opacity values for different data values via WIMP style dialog, while SIV system supports same function by sketching and the transfer function dialog is hidden. However, this induces another question: can SIV interface take place of WIMP? If the answer is yes, the next question is how to realize practical system with only SBI; if the answer is no, the next question is where is the “borderline” between SBI and WIMP.

## 6.4 Evaluation

Evaluation of usability is necessary for any HCI technology. There is no exception for SIV system. Nearly all SIV research papers emphasized the extreme importance of evaluation and stated that user study is one of their future works Owada et al. (2008); Chen et al. (2006, 2008). Although working with special researchers helps design more natural and expressive interface, few of them carry out steps of evaluation in deed, in despite of formally or informally (Akers 2006a, b; Sowell et al. 2009).

One of the obstacles is corporation. That means visualization researchers no longer develop techniques alone, but collaborate with special researcher. Even some useful paradigms exist (B dker and Gr nb k 1991; Dahlb ck et al. 1993), it is still a noticeable problem that developed SIV interfaces are lack of invaluable evaluation. Researchers should pay enough attention to evaluation of usability in bring forward new SIV technology.

One of the most difficult aspects of developing new SIV methods is evaluating their success. For many algorithms, the best measure of success is the correctness, speed and space, but that is virtually impossible to measure SIV techniques quantitatively. Some researchers revert to less direct measures, like interactivity, simplicity, minimality and visibility (Akers 2006a, b). However, do we could evaluate the usability of SIV quantitatively? Is there any benchmark?



## 7 Conclusions

This paper describes the current state of the art in sketch-based visualization, illustrates SBI has the potential to be used in visualization system through enormous diversity of techniques and applications. The terminology sketch-based, as used in this survey, has to be interpreted according, that is more loosely, for example also include invisible sketching, like free-form erasing operation.

We also introduce the SIV pipeline and divide the use of SBI in visualization into three categories and seven types. They are helpful to find which areas SBI has been used and which not. This should inspire researchers to use SBI to optimize existing methods and solve problems need proper interaction, like streamline seeding problem (McLoughlin et al. 2010; Pobitzer et al. 2011) and interactive visual analysis (Konyha et al. 2009).

There is no single interaction method could provide optimal interface for all visualization. The most appropriate method is combining several useful interaction technologies, such as brushing, multi-linked-view. We also believe that it is very helpful to make use of WIMP. The benefits of integration are twofold: one is providing practical visualization system with more interaction, the other is appropriate for novice to learn and experts to explore.

At last, we intend to highlight cooperation with domain researchers again. Without benefits stated before, cooperation could also help interaction technology designers to find what there specialists are good at or poor at. This is very important to explore a tool to leverage specialists' strengths and assist to get past their limitations. There are other challenges necessary to cope with, such as supporting uncertainty visualization, collaboration, interaction on touch displays.

In this survey, we not only want to give a snapshot of SIV but also be willing to make more emphasis on interaction in visualization. And we hope our work will inspire more research and think on the question originally: how could user handle with the data via visualization intuitively?

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